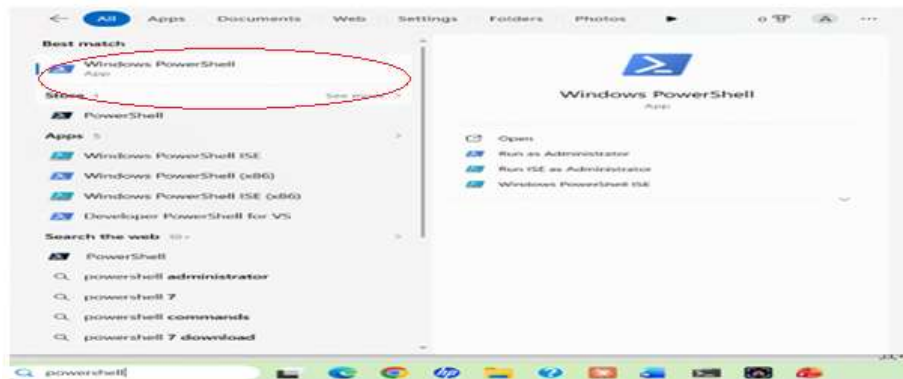
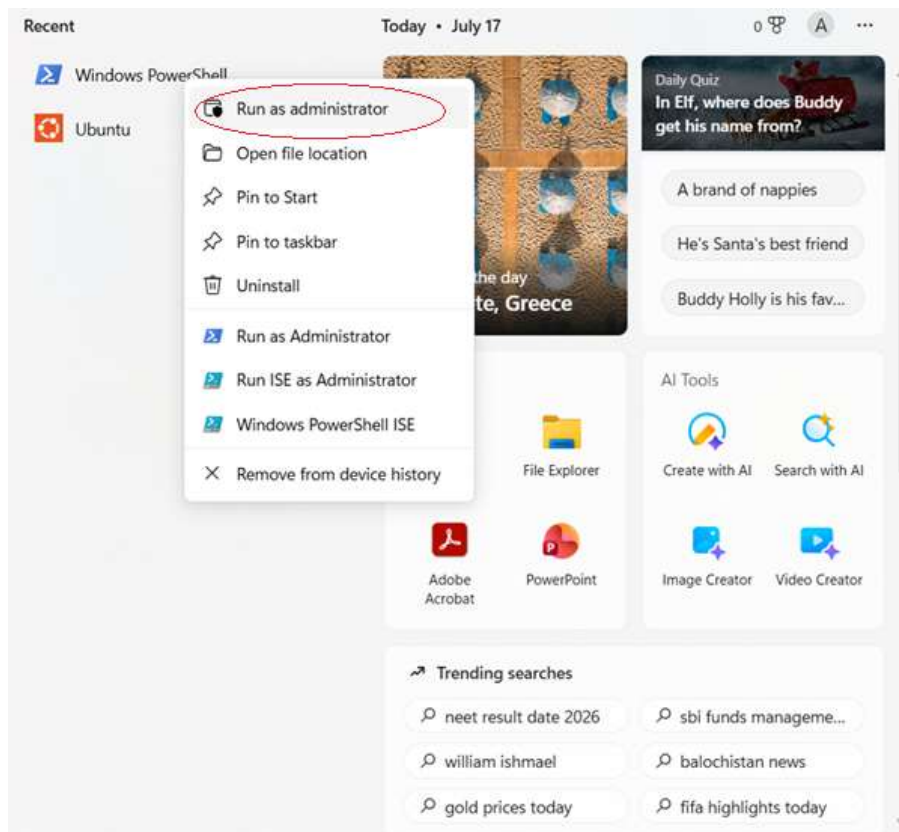


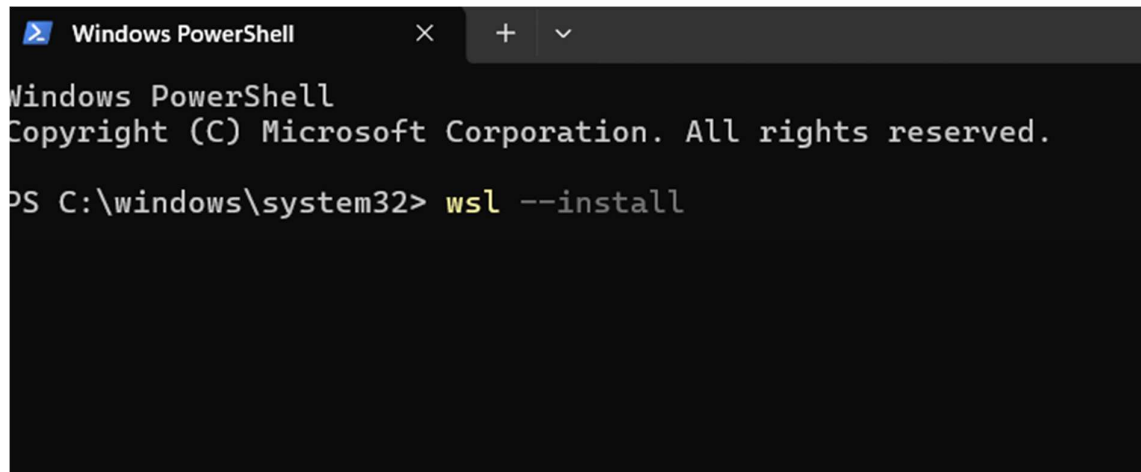
Step 1: Open PowerShell as Administrator

1. Press **Windows**.
2. Search for **PowerShell**.



3. Right-click **Windows PowerShell**.



A screenshot of a Windows PowerShell terminal window. The title bar at the top says "Windows PowerShell" with a close button (X) and a dropdown menu (v). The terminal text shows the Windows PowerShell logo, the copyright notice "Copyright (C) Microsoft Corporation. All rights reserved.", and the command prompt "PS C:\windows\system32> wsl --install".

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\windows\system32> wsl --install
```

Step 3: Restart Your Computer

After installation, restart your PC when prompted.

Step 4: Complete Ubuntu Setup

After restarting:

1. Ubuntu opens automatically (or search for **Ubuntu** in the Start menu).
2. Wait for the installation to finish.
3. Create a Linux username.

Example:

Enter new UNIX username: anusha

Step 5: Create a Password

You'll then be prompted to create a password.

New password: ****

Retype new password:****

Step 6: Update Ubuntu

```
monish_putti@monhh: ~/OSSP_Practical
monish_putti@monhh:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [417 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [97.2 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [159 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [51.2 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.2 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [349 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [67.8 kB]
Hit:12 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [492 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [118 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [97.8 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [243 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [81.0 kB]
Get:18 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [184 kB]
Get:19 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [367 kB]
Get:20 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [70.8 kB]
Get:21 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [11.3 kB]
Fetched 3170 kB in 3s (1105 kB/s)
62 packages can be upgraded. Run 'apt list --upgradable' to see them.
monish_putti@monhh:~$ sudo apt install build-essential
```

Step 7: Install GCC (C Compiler)

```
anusha_prakash@AnushaPrakash:~$ sudo apt install build-essential
Installing:
  build-essential

Installing dependencies:
  bzip2                gcc-x86-64-linux-gnu  libisl23
  cpp                  libalgorithm-diff-perl  libitm1
  cpp-15               libalgorithm-diff-xs-perl  liblsan0
  cpp-15-x86-64-linux-gnu libalgorithm-merge-perl  libmpc3
  cpp-x86-64-linux-gnu libasan8                libquadmath0
  dpkg-dev             libc-dev-bin           libstdc++-15-dev
  fakeroot             libc6-dev              libtsan2
  g++                  libcc1-0               libubsan1
  g++-15               libcrypt-dev           linux-libc-dev
  g++-15-x86-64-linux-gnu libdpkg-perl           lto-disabled-list
  g++-x86-64-linux-gnu libfakeroot            make
  gcc                  libfile-fcntllock-perl  manpages-dev
  gcc-15               libgcc-15-dev          rpcsvc-proto
  gcc-15-base          libgomp1
  gcc-15-x86-64-linux-gnu libhwasa0

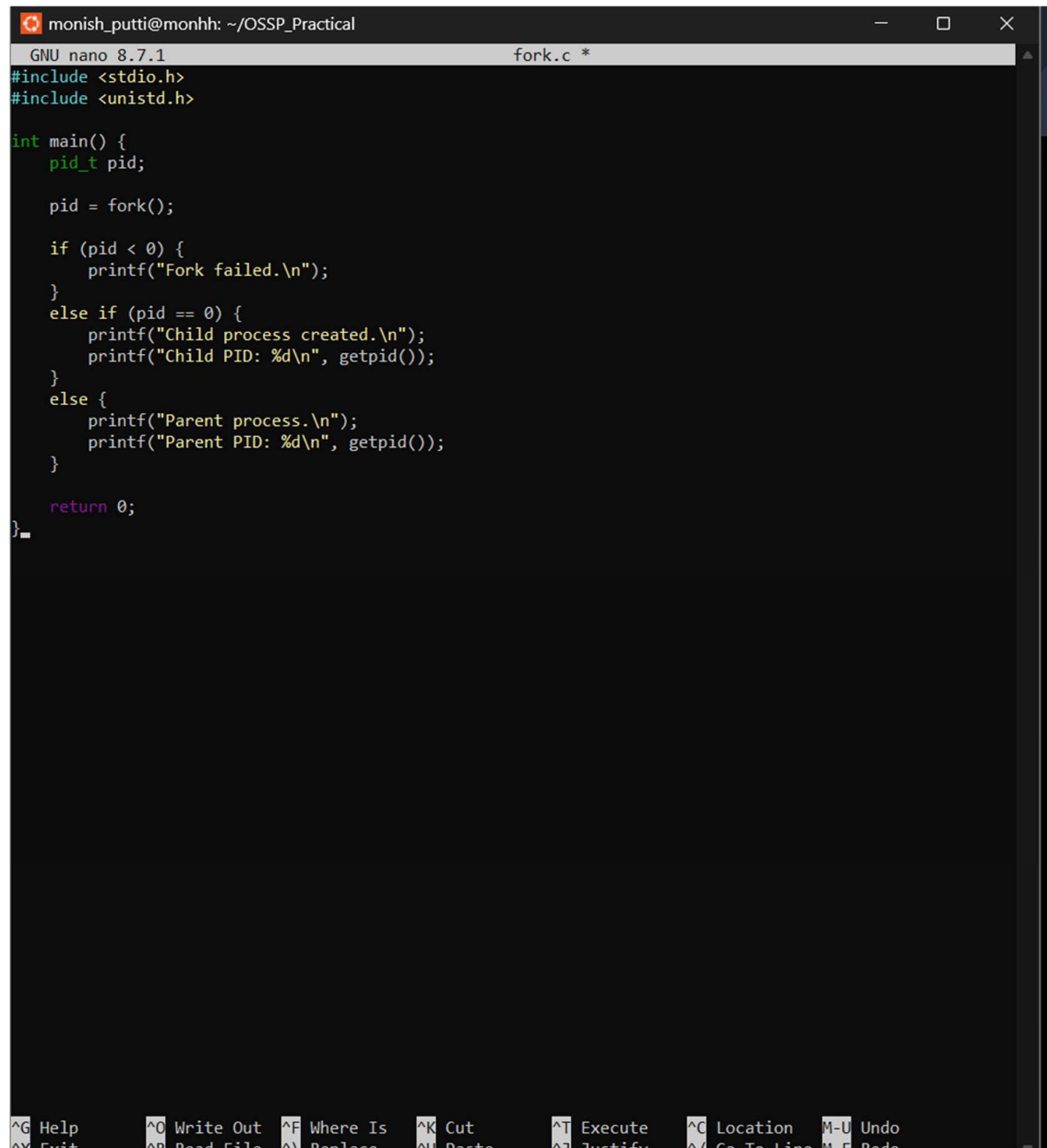
Suggested packages:
  bzip2-doc             gcc-15-doc             gcc-doc
  cpp-doc               gcc-multilib           gcc-15-multilib
```

Step 8: Verify the Installation

```
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 62
monish_putti@monhh:~/OSSP_Practical$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

Step 9: Create Your C Program

Example: nano fork.c



```
monish_putti@monhh: ~/OSSP_Practical
GNU nano 8.7.1 fork.c *
#include <stdio.h>
#include <unistd.h>

int main() {
    pid_t pid;

    pid = fork();

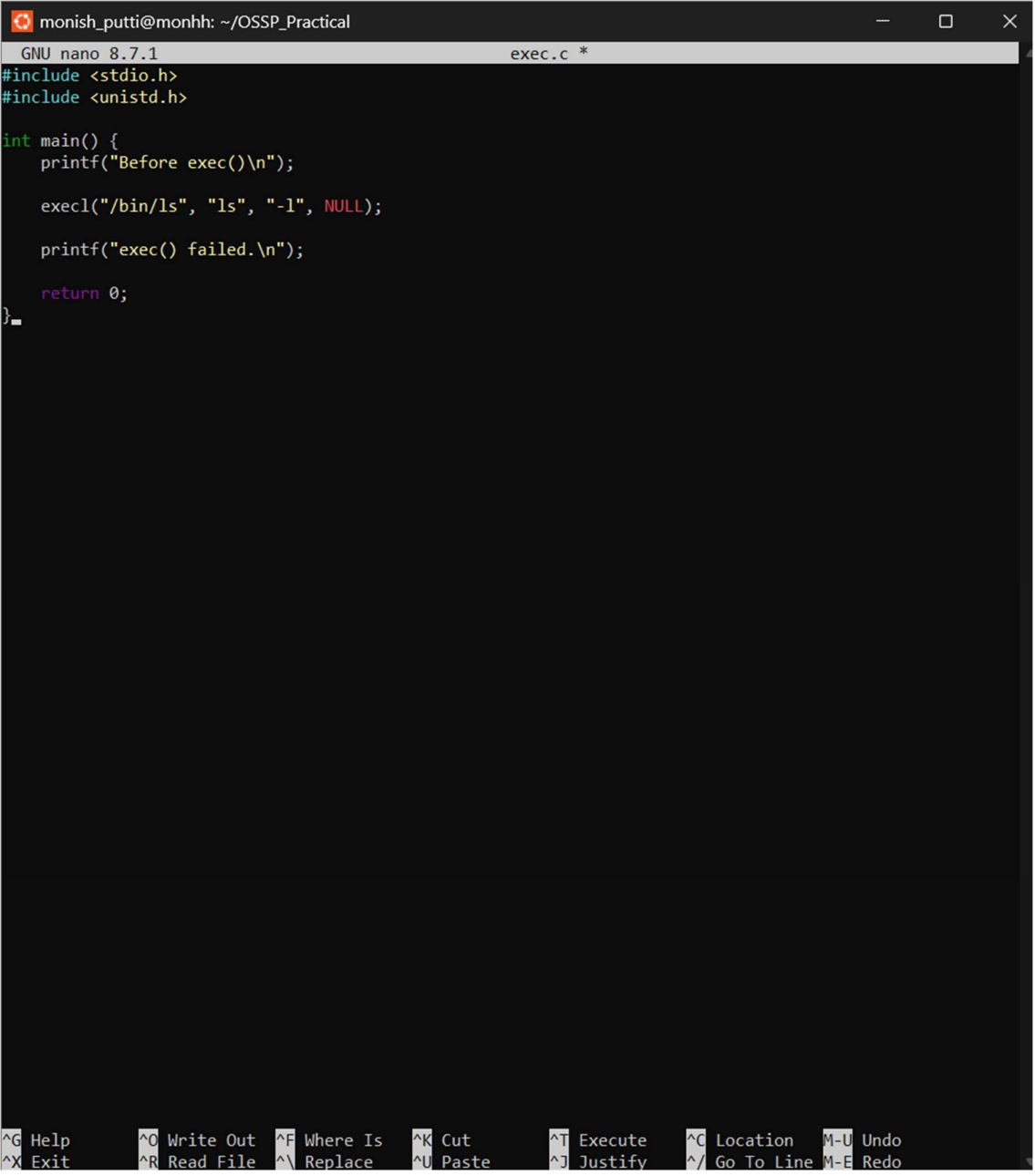
    if (pid < 0) {
        printf("Fork failed.\n");
    }
    else if (pid == 0) {
        printf("Child process created.\n");
        printf("Child PID: %d\n", getpid());
    }
    else {
        printf("Parent process.\n");
        printf("Parent PID: %d\n", getpid());
    }

    return 0;
}
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo
^X Exit ^R Read File ^_ Replace ^J Paste ^I Justify ^_ Go To Line M-F Redo

```
Child process completed.  
monish_putti@monhh:~/OSSP_Practical$ nano fork.c  
monish_putti@monhh:~/OSSP_Practical$ gcc fork.c -o fork  
./fork  
Parent process.  
Parent PID: 1131  
Child process created.  
Child PID: 1132  
monish_putti@monhh:~/OSSP_Practical$
```

Example: nano exec.c



```
monish_putti@monhh: ~/OSSP_Practical  
GNU nano 8.7.1 exec.c *  
#include <stdio.h>  
#include <unistd.h>  
  
int main() {  
    printf("Before exec()\n");  
  
    execl("/bin/ls", "ls", "-l", NULL);  
  
    printf("exec() failed.\n");  
  
    return 0;  
}
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line M-E Redo

```

monish_putti@monhh:~/OSSP_Practical$ nano exec.c
monish_putti@monhh:~/OSSP_Practical$ gcc exec.c -o exec
./exec
Before exec()
total 160
-rwxr-xr-x 1 monish_putti monish_putti 15992 Aug 24 05:18 exec
-rw-r--r-- 1 monish_putti monish_putti 175 Aug 24 05:18 exec.c
-rwxr-xr-x 1 monish_putti monish_putti 16080 Aug 24 05:16 fork
-rw-r--r-- 1 monish_putti monish_putti 384 Aug 24 05:16 fork.c
-rwxr-xr-x 1 monish_putti monish_putti 16352 Aug 24 03:13 multiple_wait
-rw-r--r-- 1 monish_putti monish_putti 1089 Aug 24 03:13 multiple_wait.c
-rwxr-xr-x 1 monish_putti monish_putti 16352 Aug 24 03:15 multiple_waitpid
-rw-r--r-- 1 monish_putti monish_putti 1169 Aug 24 03:15 multiple_waitpid.c
-rwxr-xr-x 1 monish_putti monish_putti 16168 Aug 24 05:13 parent
-rw-r--r-- 1 monish_putti monish_putti 665 Aug 24 05:13 parent.c
-rwxr-xr-x 1 monish_putti monish_putti 16168 Aug 24 05:10 process
-rw-r--r-- 1 monish_putti monish_putti 629 Aug 24 05:04 process.c
-rwxr-xr-x 1 monish_putti monish_putti 16248 Aug 24 03:19 zombie
-rw-r--r-- 1 monish_putti monish_putti 755 Aug 24 03:19 zombie.c
-rwxr-xr-x 1 monish_putti monish_putti 16256 Aug 24 03:31 zombie_fixed
-rw-r--r-- 1 monish_putti monish_putti 837 Aug 24 03:31 zombie_fixed.c
monish_putti@monhh:~/OSSP_Practical$

```

Step:10- Repository Dashboard

The screenshot shows the GitHub repository dashboard for the repository named `2520030395_OSSP` under the user `puttimonish`. The repository is public and has 7 commits. The dashboard includes a file browser on the left with files like `Practical`, `Skill`, and `README.md`. The main area displays the `README` content, which includes the repository name `2520030395_OSSP`. The right sidebar contains sections for `About`, `Releases`, `Packages`, and `Contributors`.

URL Link- https://github.com/puttimonish/2520030395_OSSP